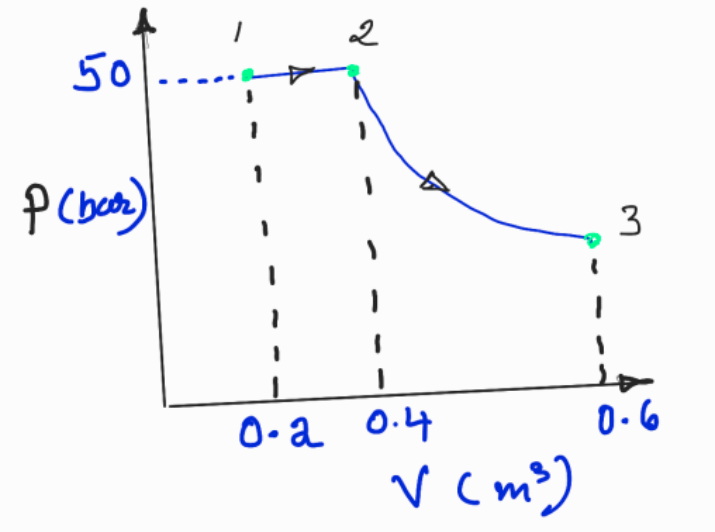
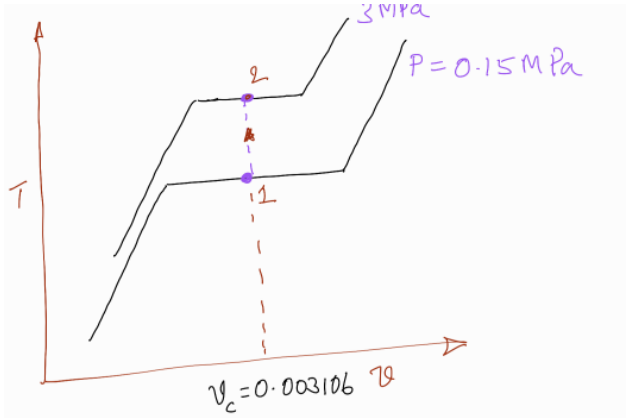
DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING MANIPAL INSTITUTE OF TECHNOLOGY, MANIPAL		
MID TERM EXAMINATION		
Subject: Engineering Thermodynamics (AAE 2123)	**Semester: III**	**Branch: Aeronautical/Automobile**
Date and Time: 13-09-2025	**Duration:90 minutes.**	**Max. Marks: 30**
Instructions: All the questions are compulsory **(Multiple choice questions carry ½ marks each)**		

Q.N	Question	Marks	CO	BT
	Multiple Choice Questions			
1	An ideal gas is contained in a rigid tank. The gas is cooled from 600 K to 300 K, and its pressure drops from 5 MPa to 2.5 MPa during the process. The specific heat at constant volume is Cv=0.72 kJ/kg-K. The work done by the gas during this process is: a) 0 kJ/kg b) −216kJ/kg c) 216kJ/kg d) More data needed Answer: Since it is constant volume process, work done is ZERO	**0.5**	**CO1**	**2**
2	Which of the following is an extensive property? a)Pressure b)Temperature c)Specific enthalpy d)None of these Answer: Extensive properties are those whose value changes with the amount of substance taken. Here pressure,temperature, specific enthalpy all will remain same.	**0.5**	**CO1**	**2**
3	Which one of the following is a **closed system**? a) Gas in a piston–cylinder with a locked piston b) Turbine c) Pump d)Boiler Answer: Closed system is one in which there is no mass transfer across the boundary of the system	**0.5**	**CO1**	**2**
4	A system undergoes two processes as shown in the P−V diagram. Process 1–2 is an isobaric expansion, and process 2–3 is a polytropic expansion. Which of the following is correct regarding the work done?	**0.5**	**CO1**	**4**

	 $W_{1-2} < W_{2-3}$ b) $W_{1-2} > W_{2-3}$ c) $W_{1-2} = W_{2-3}$ d) need more data Answer: Work done is also equal to area under the curve.			
5	The First Law of Thermodynamics for a closed system undergoing a process is expressed as: a) $\Delta U = Q - W$ b) $\Delta U = Q + W$ c) $\Delta U = W - Q$ d) $\Delta U = 0$ Answer: Heat supplied to the system – work done by the system= Change of internal energy od the system	**0.5**	**CO1**	**2**
6	A system undergoes a cyclic process and returns to its initial state. What will be the overall change in its internal energy (a)Positive (b)Negative (c)Zero (d) Depends on the process Answer: Net change of internal energy for a cyclic process is zero	**0.5**	**CO1**	**2**
7	The value of n for which a polytropic process becomes constant pressure process is: (a)0 (b)1 (c)∞ (d) γ Answer: n=1 means P=Constant	**0.5**	**CO1**	**2**
8	Gas inside a piston cylinder undergoes a process and its pressure changes from 7MPa to 3 MPa and volume changes from $0.4m^3$ to $0.8m^3$. Work done by the gas is a) Negative b) Positive c) Zero d) Need more data Answer: Volume is increasing hence work is done by the system and hence positive	**0.5**	**CO1**	**2**
9	Unit of work a) N-m	**0.5**	**CO1**	**2**

	b) J c) kJ d) all of these Answer: All are equivalent units			
10	Heat and work are forms of a) Energy of the system b) Properties of system c) Both of these d) None of these Answer: Heat and energy are two forms of energy in transaction	**0.5**	**CO1**	**2**
11	A rigid vessel containing 2 kgs of wet steam at a pressure of 0.15 MPa. When heated the states of the steam passes through the critical point. Find the following a) Volume of the vessel b) Mass of saturated water and saturated vapor present initially Steam is now heated till it reaches a pressure of 3MPa and the steam is still wet. Find the following c) Temperature of steam d) Total heat transfer from the beginning Draw the T-v diagram indicating initial and final state along with heating process Answer:  (a) Volume of the vessel, $V = v_1 m$ $= 0.003106 \times 2$ kg $V = 0.006212$ m^3 (b) (a) $P = 0.15$ MPa we have (a) (1) $v_f = 0.001053$ $v_g = 1.1594$ $\therefore v_1 = v_f + x_1(v_g - v_f)$	**4**	**CO3**	**4**

i.e $0.003106 = 0.001053 + x(1.1594 - 0.001053)$

$\Rightarrow x_1 = 0.001815$

We know $x_1 = \frac{m_g}{m_f + m_g} = \frac{m_g}{2}$

$\Rightarrow m_g = 0.001815 \times 2$

$m_g = 0.00363$

$\therefore$ $m_g = 0.00363$ kg

$m_f = 1.99637$ kg

(c) @ 3 MPa @ (2)

Temperature of steam is T_s at 3 MPa

i.e $T = 233.85^\circ C$

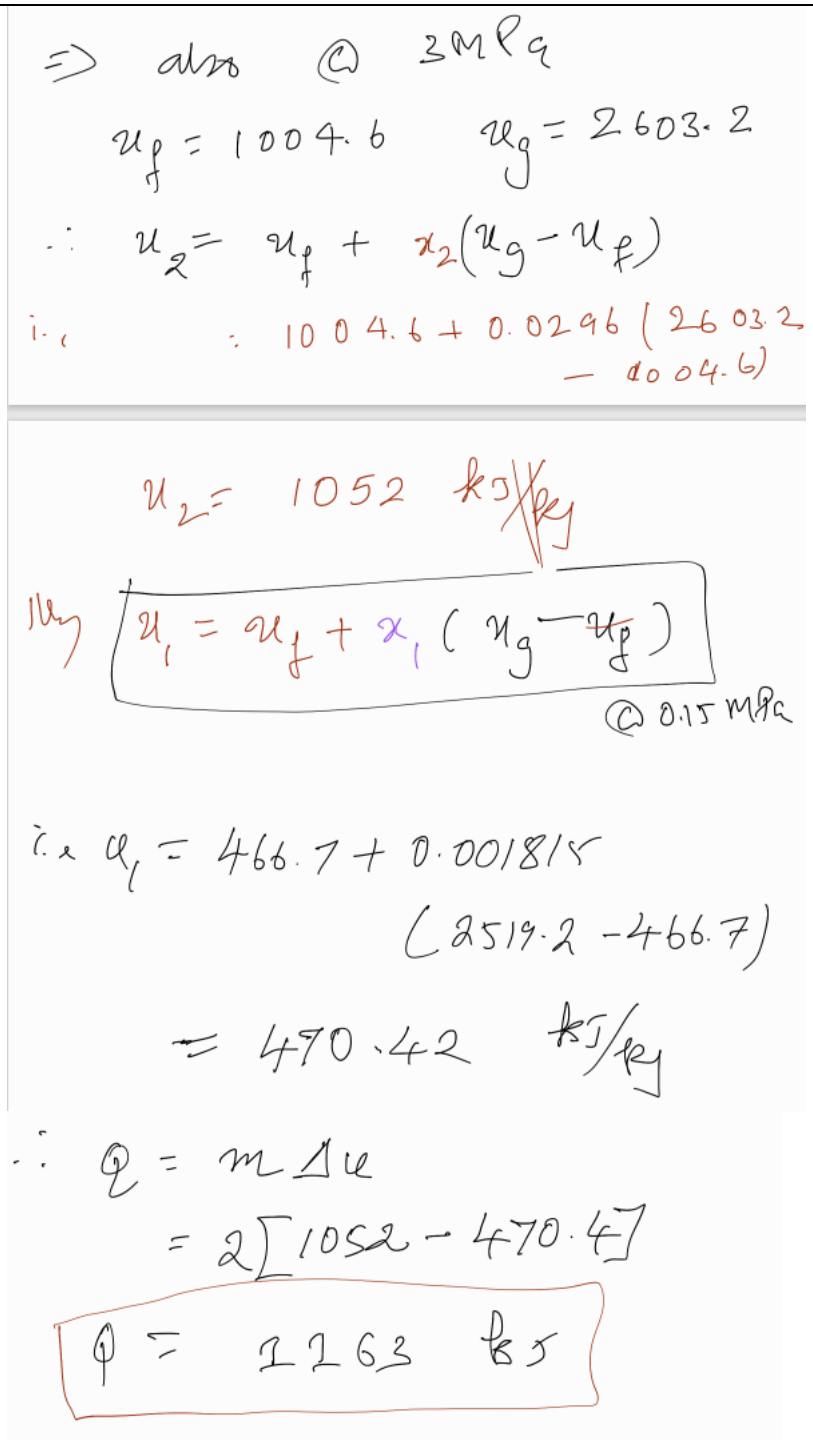
We know @ 3 MPa,

$v_f = 0.001217 \quad v_g = 0.066667$

i.e $v_2 = v_f + x_2(v_g - v_f)$

$0.003106 = 0.001217 + x_2(0.066667 - 0.001217)$

$\Rightarrow x_2 = 0.0296$

				
12	Find the properties of steam at a pressure of 2MPa and 550^{0}C. When this steam is cooled at constant pressure at what temperature it becomes saturated steam? Answer:	**3**	**CO3**	**3**

	12. @ 2 MPa Saturation Temp $T_s = 212.38^\circ C$ ∴ The given steam is superheated ∴ From superheated table @ 2MPa and 550°C $v = \frac{0.17568 + 0.19962}{2} = 0.18765\ m^3/kg$ $u = \frac{3116.2 + 3291.5}{2} = 3203.95\ kJ/kg$ $h = \frac{3468.3 + 3690.1}{2} = 3579.2\ kJ/kg$ $s = \frac{7.4337 + 7.7043}{2} = 7.569\ kJ/kgK$ If cooling is done at constant pressure it becomes saturated steam @ 212.38°C			
13	Steam enters a turbine at 1.4MPa and 300 degree Celsius expands Isentropically (reversible adiabatic) and exits as wet steam at temperature of 40 degree Celsius. What is the power out put of this turbine for a mass flow rate of 2kg/s? Neglect kinetic and potential energy changes. Answer:	**3**	**CO3**	**3**

2 We know SFEE for adiabatic and KE and PE neglected is

$$-\dot{W} = \dot{m}[(h_2 - h_1)]$$

$$\Rightarrow \dot{W} = \dot{m}[(h_1 - h_2)]$$

Given steam's $h_1 = 3040.9$ kJ/kg
[from superheated steam table]
and also $S_1 = 6.9553$ kJ/kg-K

Some isentropic

$$S_1 = S_2$$

But (2) is wet steam @ 40°C

$$\therefore \quad S_2 = S_f + x(S_g - S_f)$$

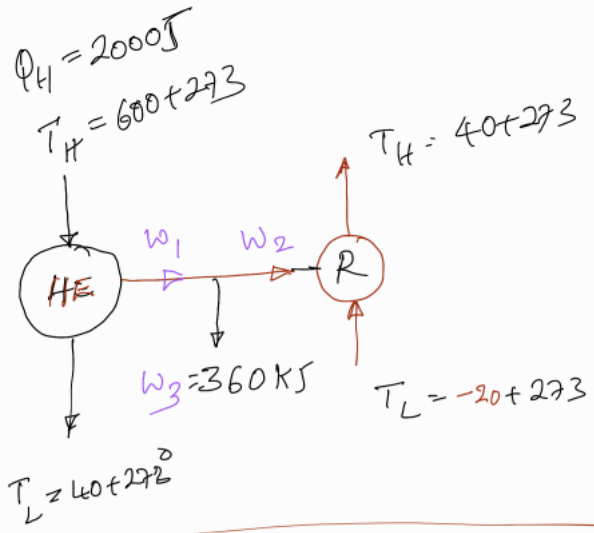
$$6.9553 = 0.5724 + x(8.2556 - 0.5724)$$

$$\Rightarrow \boxed{x = 0.83}$$

	$\therefore h_2 = h_f + x(h_g - h_f)$ @ 4° $= 167.53 + 0.83(2573.5 - 167.53)$ $h_2 = 2164.5$ kG/kg $\therefore \dot{W} = 2[3040.9 - 2164.5]$ $\dot{W} = 1752.8$ J/s $W = 1.75$ kW			
14	In a water cooling tower air enters at a height of 1 m above the ground level and leaves at a height of 7 m. The inlet and outlet velocities are 20 m/s and 30 m/s respectively. Water enters at a height of 8 m and leaves at a height of 0.8 m. The velocity of water at entry and exit are 3 m/s and 1 m/s respectively. Water temperatures are 80°C and 50°C at the entry and exit respectively. Air temperatures are 30°C and 70°C at the entry and exit respectively. The cooling tower is well insulated and a fan of 2.25 kW drives the air through the cooler. Find the amount of air per second required for 1 kg/s of water flow. The values of c_p of air and water are 1.005 and 4.187 kJ/kg K respectively Answer:	**3**	**CO2**	**4**

	14) 2.25 kW $\dot{Q} - \dot{W} = \dot{m}_{air}\left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right]$ $\Delta h_{air} = 1.005 \times 10^3 (70-30)$ $= 40.3 \times 10^3$ J/kg $\dot{Q} = 1 \times 4.187 \times 10^3 (80-50)$ $= 125.6$ kW $\therefore 125.6 \times 10^3 - (-2.25 \times 10^3) = \dot{m}\left[40.2 \times 10^3 + \frac{30^2 - 20^2}{2} + 9.81(7-1)\right]$ $\Rightarrow \dot{m} = 3.16$ kg/sec			
15	A reversible heat engine operates between two reservoirs at temperatures of 600°C and 40°C. The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 40°C and –20°C. The heat transfer to the heat engine is 2000 kJ and the net work output of the combined engine–refrigerator plant is 360 kJ. Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C. **Answer:**	**3**	**CO2**	**3**

15.



$$\eta_H = \frac{T_H - T_L}{T_H} = \frac{873 - 313}{873} = 0.64$$

$$W_1 = 0.64 \times 2000 = 1283 \text{ kJ}$$

∴ Heat rejected by HE

$= 2000 - 1283$

$$Q_{L_{HE}} = 717 \text{ kJ}$$

We know

$$W_2 = 1283 - W_3$$

$$W_2 = 1283 - 360 = 923\ kJ$$

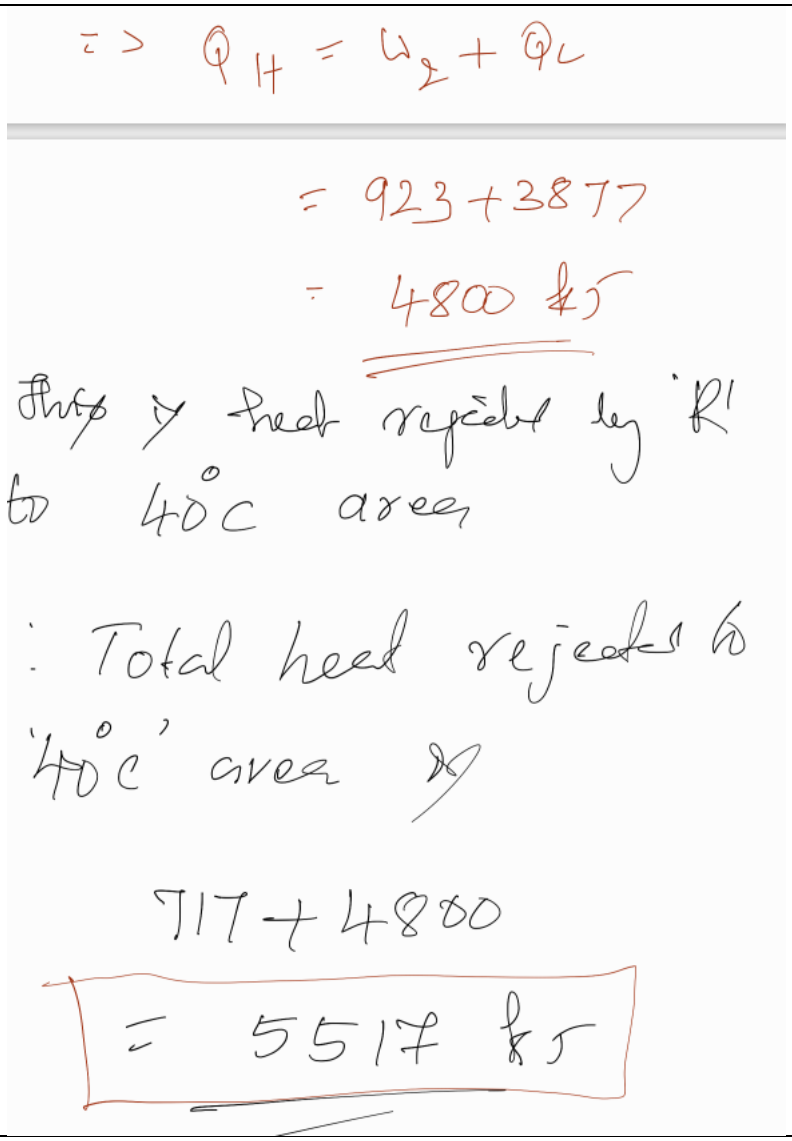
$$COP_R = \frac{T_L}{T_H - T_L} = \frac{253}{313 - 253} = 4.2$$

Also $COP_R = \frac{Q_L}{W_2}$

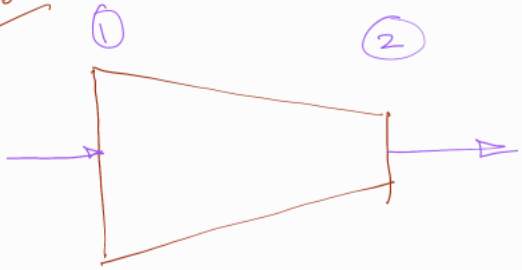
$$\Rightarrow Q_L = COP_R \times W_2$$

$$= 4.2 \times 923$$

$$= 3877\ kJ$$

				
16	A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exit from the nozzle. (b) If the inlet area is 0.1 m^2 and the specific volume at inlet is 0.187 m^3/kg, find the mass flow rate. (c) If the specific volume at the nozzle exit is 0.498 m^3/kg, find the exit area of the nozzle. **Answer:**	**3**	**CO2**	**3**

16



$h_1 = 3000$ kJ/kg $h_2 = 2762$ kJ/kg

$V_1 = 60$ $V_2 = ?$

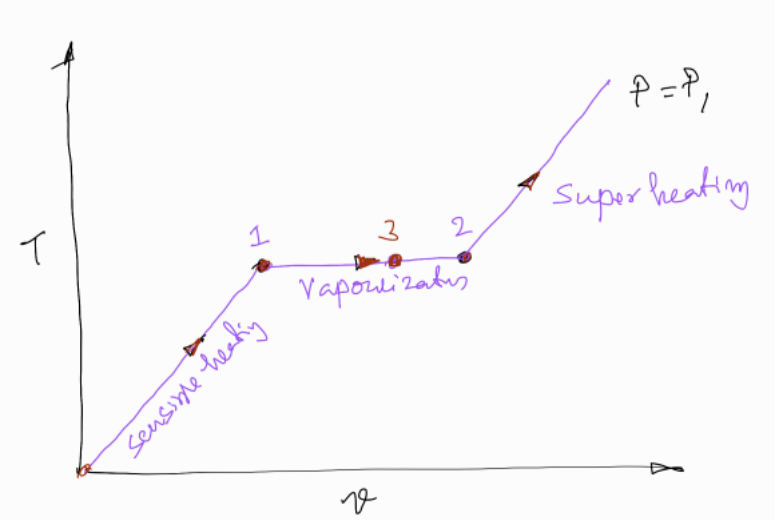
$$\dot{Q} - \dot{W} = \dot{m}\left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2}\right]$$

$$(h_1 - h_2) = \frac{V_2^2 - V_1^2}{2}$$

$$\Rightarrow V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$$

$$\boxed{V_2 = 693 \text{ m/s}}$$

	mass flow rate $\dot{m}_1 = A_1 V_1 \cdot \frac{1}{v_1}$ $= A_1 \times V_1 \times \frac{1}{v_1}$ $= 0.1 \times 60 \times \frac{1}{0.187}$ $\dot{m}_1 = 32 \text{ kg/s}$ $\dot{m}_1 = \dot{m}_2$ $32 = A_2 \times V_2 \times \frac{1}{v_2}$ $32 = A_2 \times 693 \times \frac{1}{0.498}$ $\Rightarrow A_2 = 0.0229$ $\approx 0.023 \text{ m}$			
17	Draw a neat T-v diagram of phase change process of water indicating all salient points. Answer:	**2**	**CO3**	**2**

	 1 – Saturated liquid 2 – Saturated vapour 3 – Wet Steam			
18	A mass of 1.5 kg of air (density =1.16kg/m^3)is compressed inside a piston cylinder from 0.1MPa to 0.7 MPa. The process of compression follows Pv=C. How much work done in this process? During this process a heat of 150kJ is lost from the cylinder to the surrounding. What is the change of internal energy of the air? Answer:	**2**	**CO2**	**3**

$m = 1.5$ kg, $\rho = 1.16$, $v = \frac{1}{1.16}$

$V = v \cdot m$

$P_1 = 0.1$ MPa $\quad P_2 = 0.7$ MPa

gives $Pv = C$ means Isothermal process $[\because n = 1]$

Work done in isothermal proc

$$W = P_1 V_1 \ln\left(\frac{P_1}{P_2}\right) \text{ Joules}$$

$$= 0.1 \times 10^6 (v \times m) \ln\left[\frac{0.1}{0.7}\right]$$

$$= 0.1 \times 10^6 \left(\frac{1.5}{1.16}\right) \ln\left(\frac{0.1}{0.7}\right)$$

$W = -251.6$ kJ

We know

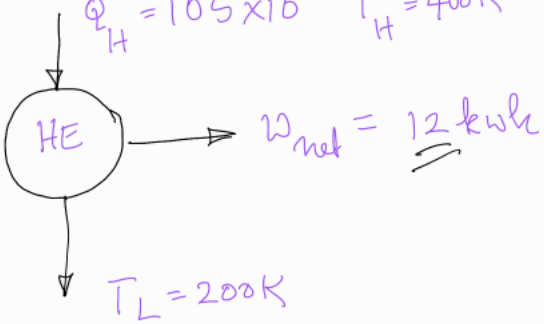
$$Q - W = \Delta U$$

$$\therefore \quad -150 \times 10^3 - (-251.6 \times 10^3) = \Delta U$$

$\Delta U = 101.6$ kJ

19	An inventor claims to have developed an engine that takes in 105 MJ at a temperature of 400 K, rejects heat to temperature of 200 K, and delivers 12 kWh of mechanical work. Would you advise investing money to put this engine in the market? Answer:	**2**	**CO2**	**3**

19



$$\eta_{HE} = \frac{T_H - T_L}{T_H} = \frac{400 - 200}{400} = 50\%$$

it means out of 105×10^6 Joules

Supplied maximum 52.5×10^6 Joules

would be available as work.

In this case work available

is 12 kwh $= 12 \times 10^3 \times 3600$

$= 43.2 \times 10^6$ Joules

It means his claim is still valid and one can invest